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Cisco Secure Workload — TSA Pipeline Security

Reference Design | Owners and Operators of TSA-Designated Natural Gas / Hazardous Liquid Pipelines

Version: 1.0 (draft — pending SME review) | **Standard:** TSA Security Directive Pipeline-2021-02 series (current revision; the SD has been republished as 02A, 02B, 02C, and 02D as TSA has updated the requirements; this reference design is generic to the current effective version) and Security Directive Pipeline-2021-01 series for incident reporting | **Audience:** Pipeline owner/operators designated by TSA as Critical Pipeline owner/operators, and the cybersecurity teams supporting them | **Environment:** IT estate supporting OT (corporate IT, engineering workstations, SCADA jump servers, historian/MES, vendor-access servers — *not* the pipeline OT control devices themselves)

Reader's Guide

Who this is for. TSA-designated owner/operators of natural gas and oil/hazardous liquid pipelines and their cybersecurity, IT, and OT operations teams. The reference design is also useful for pipeline operators not currently TSA-designated who want to align proactively with the SD framework.

Scope boundary you must understand before reading further. Cisco Secure Workload (CSW) is a workload-protection and segmentation platform for **IT estate** — servers, virtual machines, containers, and cloud workloads. CSW does **not** enforce policy on RTUs, flow computers, PLCs, valve actuators, or any other Level 0–2 OT device on the pipeline itself. CSW's TSA SD contribution is at the **IT-side of the IT/OT boundary** and within the IT systems that support pipeline operations — engineering workstations, SCADA jump servers, historian databases, MES/EAM systems, vendor remote-access hosts, identity/PKI services, and the corporate IT systems that interact with Critical Cyber Systems on the IT side.

Questions this reference design helps you answer:

- *Section II — Critical Cyber System identification: do I have a current inventory of every IT system that supports pipeline operations, with the supporting Critical Cyber System linkage documented and continuously refreshed?*
- *Section III.A — Network Segmentation (IT/OT separation): can I prove that segmentation between IT and OT is enforced at the workload level on the IT side, and that the documented IT-to-OT data flows are the only flows actually occurring?*

- *Section III.B — Access Control Measures: can I produce evidence that access to Critical Cyber Systems and to the IT systems that support them is least-privilege, monitored, and reviewed on a documented cadence?*
- *Section III.C — Continuous Monitoring and Detection: when an anomalous IT-side flow appears (e.g., a corporate workstation starts talking to a SCADA jump server it has never touched before), is it detected in time to investigate?*
- *Section III.D — Reduce Risk of Exploitation of Unpatched Systems: can I produce a per-IT-workload vulnerability inventory with remediation tracking, prioritised by which workloads support Critical Cyber Systems?*
- *Cybersecurity Incident Response Plan: when a Cybersecurity Incident occurs on the IT estate, can I reconstruct the process and flow context to support TSA's 24-hour reporting requirement?*
- *Cybersecurity Assessment Plan (CAP): can I produce evidence of annual implementation assessment for the IT-side controls, with measurable test outcomes?*

What you'll need. Your current Critical Cyber System inventory (per Section II of the SD), your most recent Cybersecurity Incident Response Plan, your Cybersecurity Architecture Design Review documentation, and your current CAP. CSW will augment these governance artefacts with workload-level evidence; it will not replace them.

Where to start. Section 2 if you are scoping CSW relative to the Critical Cyber System inventory; sections 3–4 if IT/OT segmentation is the priority; section 5 if access control evidence is the gap; section 6 if the CAP requires improved continuous monitoring evidence; section 9 if vendor and third-party visibility is the open item.

Important. TSA SDs are issued under TSA's regulatory authority; designated owner/operators are required to comply. Nothing in this reference design substitutes for the responsible Senior Cybersecurity Officer's accountability under the SD or for the formal TSA inspection response. CSW gives the responsible officer technical evidence inputs for the operator's compliance programme; it does not substitute for the designated security programme.

CSW primer — if you are new to Cisco Secure Workload

Cisco Secure Workload (CSW) is a **workload protection platform**. A lightweight **agent** on each server/VM/container observes processes and network flows; **cloud connectors** add AWS/Azure/GCP inventory where agents are not deployed.

CSW term	Meaning	Why compliance teams care
Scope	Logical boundary (CDE, PHI zone, CUI enclave)	Defines the systems you must prove are isolated
Label / filter	Tag or query assigning workloads to scopes	Automates scope membership — reduces drift
ADM	Application Dependency	Live network/data-flow

CSW term	Meaning	Why compliance teams care
	Mapping from observed traffic	diagram for assessors
Workspace	Policy container for one scope or application	Where allow/deny rules are authored
Monitor → Simulation → Enforce	Safe rollout sequence	Simulation = change-board evidence before blocking traffic
Denied Connections	Log of flows blocked by policy	Primary proof that enforcement operates

Console areas: Investigate (inventory, flows, vulns) · Defend/Segmentation (policy) · Manage (agents) · Platform (connectors) · Administration (audit log)

Read next: [Compliance evidence playbook](#) (full step-by-step) · [About CSW](#) (platform intro)

Universal evidence workflow

Execute these phases for **this framework's** compliance boundary. Map exports to control IDs in the **Reporting & Evidence** section below.

Phase	Goal	Key CSW actions	Typical duration
1 — Coverage	Every in-scope workload in CSW	Install agents/connectors ; apply compliance labels; create scope	Days 1–10
2 — Baseline	Machine-generated flow map	Run ADM ≥2 weeks; export clusters; app-owner signoff	Days 11–28
3 — Policy	Designed isolation before enforce	Build workspace; Simulation mode; fix false positives	Days 29–45
4 — Operate	Continuous proof between audits	Enforce; quarterly export pack; ADM refresh every 90 days	Ongoing

Phase 1 checklist (coverage)

- ☐ In-scope host list reconciled to CSW Inventory (100% or documented exceptions)

- ☐ Labels applied: `compliance:<framework>`, `data:<classification>`, `env:<tier>`
- ☐ CSW scope created matching compliance boundary
- ☐ **Export:** inventory CSV + agent status screenshot

Phase 2 checklist (baseline)

- ☐ ADM running on compliance scope for full business cycle (≥ 2 weeks)
- ☐ Unexpected flows documented (shadow IT, vendor egress, scope creep)
- ☐ App owners signed cluster-to-application mapping
- ☐ **Export:** ADM diagram + flow samples with process context

Phase 3 checklist (policy)

- ☐ Default-deny posture defined for sensitive scope
- ☐ ADM-imported rules refined; Simulation run ≥ 1 week
- ☐ Change tickets for false-positive fixes; exception register updated
- ☐ **Export:** policy export + simulation report

Phase 4 checklist (operate)

- ☐ Enforcement enabled on pilot scope; negative test recorded in Denied Connections
- ☐ Quarterly pack: inventory, policy, denies, vulns, audit log (see playbook)
- ☐ SIEM integration verified (sample events)
- ☐ **Export:** enforcement screenshot + quarterly binder

What CSW evidence does not replace

Physical access, HR/training records, encryption key management, signed BAAs/vendor contracts, formal pen tests, and assessor attestation still require separate programmes. CSW addresses the **workload-resident** slice: segmentation, flows, process context, vuln reachability, and change drift.

CSW effectiveness for this framework

- IT/OT segmentation evidence (Section III.A) on IT workloads
- Access control + monitoring (III.B/C) at workload tier
- CIRP/CAP evidence packs from forensic flow export

Compared to manual programmes: static diagrams and annual firewall samples age immediately; CSW ties evidence to **live workload behaviour** and produces queryable exports on demand — supporting "operating effectively" language in PCI v4.0, SOC 2 CC7, and HIPAA risk analysis.

1. Overview

The TSA Pipeline Security Directive 2021-02 series (currently in revision; consult the TSA web site for the active revision letter) imposes mandatory cybersecurity requirements on TSA-designated Critical Pipeline owner/operators in the United States. The directive mirrors much of the CISA CPGs and NIST 800-53 control family in content, but applies on a sector-specific basis with TSA enforcement.

The SD's substantive cybersecurity requirements fall under four broad sections:

Section	Topic	CSW relevance
Section II	Critical Cyber System identification + Cybersecurity Coordinator + governance	Inventory + scope labelling on the IT side
Section III.A	Network Segmentation (IT/OT separation)	Direct — IT-side segmentation up to the IT/OT boundary
Section III.B	Access Control Measures	Direct — workload-level least-privilege; deny-by-default for cross-zone access
Section III.C	Continuous Monitoring & Detection	Direct — behavioural detection + flow telemetry to SIEM
Section III.D	Reduce Risk of Exploitation of Unpatched Systems	Direct — continuous CVE inventory tied to workload context
Cybersecurity Incident Response Plan	Plan, test, report	Forensic flow + process telemetry for incident reconstruction
Cybersecurity Assessment Plan (CAP)	Annual implementation assessment	Evidence pack supporting the CAP self-assessment
Cybersecurity Architecture Design Review	Architecture + design assessment	Workload-level inventory and dependency graph as inputs

Where CSW fits — the short version. TSA SDs explicitly call for network segmentation between IT and OT, continuous monitoring, access control, and reduction of unpatched-system risk. CSW is a direct fit for the **IT side** of all four substantive III.A–III.D requirements. CSW supplements (does not replace) your OT-aware monitoring and your IT/OT firewall pair at the boundary.

Where CSW does not fit. CSW does not enforce policy on RTUs, flow computers, PLCs, IEDs, HMIs at the OT layer; does not perform deep packet inspection of OT protocols (DNP3, IEC 61850, Modbus, OPC-UA); is not certified as a Cybersecurity Architecture Design Review tool; and does not author the Cybersecurity Incident Response Plan. Your OT-aware monitoring stack (Cisco Cyber Vision, Claroty, Nozomi, Dragos, etc.) and your boundary firewall remain the controlling artefacts at and below the IT/OT boundary.

2. Section II — Critical Cyber System Identification

The SD requires identification of every Critical Cyber System (those that, if compromised or rendered unavailable, would result in operational disruption). The inventory must be kept current and made available to TSA on request.

CSW does not perform Critical Cyber System designation. That is an engineering and risk decision driven by the operational impact analysis your Senior Cybersecurity Officer signs off. CSW *does* host the labels that downstream IT-side controls depend on.

CSW Implementation:

Step 1: Receive the Critical Cyber System list from the operations risk function

→ Per IT workload supporting a CCS:

```
ccs_id:      [reference to your CCS register entry]
ccs_function: [scada-jump | historian | mes | hmi-server-it |
               vendor-access | engineering-workstation |
               identity | patch-repo | other-it-supporting]
ot_facing:   true | false   (does the workload reach into OT?)
site:        [pipeline segment / facility name]
```

Step 2: Apply consistent labels to every CSW-managed workload

CSW UI → Inventory → Bulk Label

Mandatory labels:

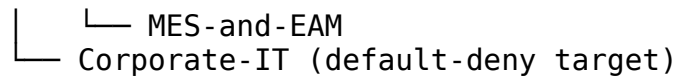
ccs_id, ccs_function, ot_facing, site, region, classification

Step 3: Build CSW Scopes per (region, site, ccs_function)

CSW UI → Organize → Scopes → New

→ Recommended hierarchy:

```
Root
├── Pipeline-IT-Estate
│   ├── OT-Facing (ot_facing=true)
│   │   ├── SCADA-Jump-Hosts
│   │   ├── Historians
│   │   ├── HMI-Servers-IT
│   │   ├── Engineering-Workstations
│   │   └── Vendor-Access-Hosts
│   └── CCS-Supporting-IT (CCS but not ot-facing)
│       ├── Identity-and-PKI
│       └── Patch-Repositories
```



Step 4: Validate scope membership against the CCS register
→ Cross-check CSW count vs. the CCS register; reconcile any delta with the CCS Asset Owner before formal sign-off

Evidence: Per-scope membership snapshot (CSV) with workload identity, OS, owner, CCS reference, OT-facing flag, site, and last- seen timestamp. This becomes the IT-side companion to your Section II Critical Cyber System inventory.

3. Section III.A — Network Segmentation (IT/OT Separation)

TSA SD Section III.A is the foundational segmentation requirement. The directive calls for network segmentation policies and controls ensuring that the OT system can continue to safely operate in the event of an IT system compromise, *and vice versa*.

Architectural reality. Your IT/OT boundary firewall(s) remain the controlling artefact for the actual IT/OT boundary. CSW contributes by ensuring *the IT-side itself is segmented*, so that a compromised corporate-IT workload cannot reach the boundary firewall from an undocumented direction.

CSW Implementation — IT-side segmentation pattern:

Step 1: Define the IT-side OT-adjacent enclave policy workspace
CSW UI → Defend → Segmentation → New Workspace
Name: tsa-pipeline-it-side-{site}
Scope: site = {site} AND ot_facing = true OR ccs_function != null

Step 2: Compute the as-observed baseline via ADM
Run ADM ≥ 14 days; capture maintenance and turnaround windows
Reject "noise" clusters; rename clusters to operational roles

Step 3: Layer SD-aligned hardening rules on top of the baseline
DENY: Corporate-IT → OT-Facing-IT
(only documented hosts may reach the OT-adjacent IT tier)
DENY: Corporate-IT → Boundary-Firewall-IPs
(the IT/OT firewall itself only accessible from documented management network)
DENY: OT-Facing-IT → Internet
(no direct internet from a SCADA jump host or historian)
ALLOW: Vendor-Access-Hosts → vendor-allowlisted FQDNs only
(vendor remote support path documented and bounded)
ALLOW: Patch-Repositories → vendor patch URLs only
(so patching can occur without broad internet)
LOG: Any unmatched flow inbound to OT-Facing-IT or CCS-Supporting-IT
(becomes a Section III.C continuous-monitoring candidate)

Step 4: Run in Simulation for ≥ 30 days

- Resolve every legitimate-but-blocked flow with the asset owner; document each exception in the workspace notes for the audit trail referenced from the SD.

Step 5: Promote to Enforce; gate by change-management approval

- Each subsequent change captured in CSW change history; this is the IT-side complement to your Section III.A documentation.

Evidence: (a) workspace export per site; (b) simulation → enforcement transition log; (c) quarterly enforcement diff report delivered to the responsible Senior Cybersecurity Officer.

4. Section III.A continued — Documented Data Flows

The SD calls for documentation of permitted communication between IT and OT systems, with denial of all other communication.

CSW Implementation:

Step 1: Enumerate observed IT→OT flows

CSW UI → Investigate → Flow Search

Filter: `source_scope c Pipeline-IT-Estate AND destination_subnet ∈ OT-network-ranges`

- Output: observed IT→OT flow inventory per site for instrumented IT workloads and configured OT destination definitions

Step 2: Reconcile against the documented flows in your Cybersecurity Architecture Design Review

- For each observed flow, find the corresponding documented flow; flag missing flows (observed but not documented) and stale documentation (documented but no longer observed)

Step 3: Propose policy update + documentation update for each delta

- Observed-but-undocumented: investigate, then either remove the flow or document the rationale
- Documented-but-not-observed: candidate for removal during the next architecture review cycle

Step 4: Continuously monitor for new IT→OT edges

- CSW alert: "new flow source/destination pair to OT range"
- Routes to the IT/OT change board for review under Section III.A

Evidence: (a) IT → OT flow inventory dated within the period; (b) observed-vs-documented delta report; (c) new-edge alert log.

5. Section III.B — Access Control Measures

The SD requires access control measures to enforce least privilege and to prevent unauthorised access to Critical Cyber Systems.

CSW Implementation — Workload-level least-privilege:

Step 1: Enforce deny-by-default at the CSW workload identity layer

- Every flow must match an explicit allow rule; everything else is denied with audit logging
- Identity for CSW policy = workload + scope + tag set, not just IP or subnet (this matters because IPs reassign in cloud and virtualised environments)

Step 2: Limit lateral access between CCS-Supporting-IT workloads

- Default: workloads in different ccs_function categories may not communicate
- Explicit allow only for documented operational paths (e.g., historian needs read from SCADA-jump for replication)

Step 3: Limit access to vendor-access hosts to specific source set

- ALLOW: {documented MFA-VPN pool} → vendor-access hosts only
- DENY: any other source → vendor-access hosts

Step 4: Schedule cadence-based access review

- Quarterly: review the allow ruleset; sunset any allow rule whose last-observed flow is older than 90 days
- Annual: full ruleset review with the Cybersecurity Officer

Evidence: (a) policy export with rule provenance and last-flow timestamp; (b) quarterly access review minutes; (c) sunset log for removed rules.

6. Section III.C — Continuous Monitoring and Detection

The SD calls for continuous monitoring and detection policies and procedures designed to detect cybersecurity threats and correct anomalies that affect Critical Cyber System operations.

CSW Implementation:

Step 1: Enable behavioural / process-based detections on CSW-managed workloads

CSW UI → Defend → Forensics → Rules

- Out-of-the-box rules: privilege escalation, persistence, lateral movement, suspicious child processes
- Custom rules: encode SD-relevant abuse patterns
 - Example: "Process on SCADA-jump host writes to startup paths"
 - Example: "Outbound to non-allowlisted FQDN from historian"
 - Example: "New listening port opened on engineering workstation"

Step 2: Conversation-graph anomaly detection

- CSW auto-flags new edges in the dependency graph for OT-Facing-IT and CCS-Supporting-IT scopes
- Auto-flags changes in port/protocol distribution per workload

Step 3: Wire alerts to the SOC

- CSW UI → Manage → Data Tap → SIEM
- Splunk, QRadar, Sentinel, Cisco XDR
- Tag every alert with ccs_id and ot_facing so the SOC can route by criticality
- Pair with your OT-aware sensor alerts for full IT+OT coverage

Step 4: Track detection latency

- CSW timestamps every detection; SIEM timestamps acknowledgement
- Compute MTTD per CCS function; report quarterly to the Senior Cybersecurity Officer

Evidence: (a) detection rule catalogue with last-modified dates; (b) SIEM forwarding configuration; (c) MTTD trend report per CCS function.

7. Section III.D — Reduce Risk of Exploitation of Unpatched Systems

The SD requires a documented patch management strategy and the controls to reduce risk on unpatched IT systems.

CSW Implementation:

Step 1: Continuous CVE inventory at workload level

- CSW UI → Investigate → Vulnerability Report
- Per-workload CVE list with CVSS, exploit availability, EPSS score
- Filter by ccs_function != null → prioritise CCS-supporting tier

Step 2: Compensating controls for unpatchable systems

- For workloads where patching is delayed (vendor lifecycle, operational window, regulatory test):
 - Restrict source set for the vulnerable port
 - Add behavioural detection for known exploit patterns
 - Log every flow involving the affected port for review
- Document each compensating control in the workload's CSW workspace notes; this is the audit trail for the "reduce risk" standard when patching is not feasible

Step 3: Patch verification

- After patch, CSW package inventory updates within the next snapshot cycle
- Diff report shows the patched version; CVE inventory drops the addressed vulnerability automatically

Step 4: Reporting

- Monthly: CVE backlog by CCS function and severity
- Quarterly: trend of mean-time-to-patch by impact rating
- Annual: input to the CAP self-assessment

Evidence: (a) per-CCS CVE list dated to the period; (b) patch remediation tracking with ticket IDs; (c) compensating control register for unpatched-but-mitigated workloads.

8. Cybersecurity Incident Response Plan

The SD requires owner/operators to develop and maintain a Cybersecurity Incident Response Plan and to report Cybersecurity Incidents to CISA within 24 hours.

CSW Implementation — Incident Reconstruction Bundle:

For each Cybersecurity Incident on a CSW-managed workload, assemble:

- (a) Affected workload set
 - Inventory query: hostnames, IPs, OS, CCS labels, site, ot_facing flag
- (b) Time-anchored flow record
 - CSW historical conversations: source, destination, port, byte counts, timestamps, process attribution
- (c) Process activity record
 - Every process spawned on affected workloads in the window
- (d) Software footprint at time of incident
 - Package list, CVE exposure at that moment (not "today")
- (e) Containment evidence
 - Quarantine policy applied; flow drop count after policy; time-to-quarantine
- (f) Configuration baseline diff
 - Before/after package list, listening ports, process set

These six artefacts populate the incident dossier your IRP requires and the CISA reporting submission you must make within 24 hours under the SD.

Evidence: (a) standard incident dossier template; (b) MTTD/MTTR metrics by CCS function; (c) annual aggregate incident-trend report for the Senior Cybersecurity Officer.

9. Cybersecurity Assessment Plan (CAP)

The SD requires owner/operators to develop and maintain a CAP and to conduct annual assessments. CSW directly supports the IT-side evidence layer for the CAP.

CSW Implementation — CAP evidence pack:

Annual CAP self-assessment evidence (per IT-side substantive control):

Section III.A (segmentation):

- Workspace export per site; simulation→enforcement log; observed-vs-documented IT→OT flow delta

Section III.B (access control):

- Policy export with rule provenance; quarterly review minutes; sunset log for removed rules

Section III.C (continuous monitoring):

- Detection rule catalogue; SIEM forwarding configuration; MTTD trend report; sample alert with full forensic context

Section III.D (vulnerability/patching):

- CVE inventory dated to assessment period; patch tracking with ticket IDs; compensating control register for delayed patches

Cross-cutting:

- Inventory snapshot dated to assessment start
- Configuration baseline diff log for the period
- Quarterly diff reports demonstrating continuous improvement

Evidence: (a) annual CAP evidence binder organised by SD section; (b) prior-period delta showing what changed since last CAP; (c) Senior Cybersecurity Officer sign-off summary.

10. Cybersecurity Architecture Design Review

The SD requires periodic architecture design reviews. CSW does not *perform* the review, but provides telemetry-based inputs.

CSW inputs to the design review:

1. Current IT-side workload inventory with CCS labelling.
2. Current ADM-derived dependency graph per CCS function.
3. Observed IT → OT flow inventory for reconciliation against the documented architecture.
4. Vendor-egress inventory per OT-Facing-IT workload.
5. Quarterly enforcement diff history showing how the IT-side has evolved since the last review.

These inputs let the architecture review focus on *gaps and changes* rather than rebuilding the as-is picture from scratch.

11. Mapping Table — TSA SD Requirement → CSW Capability

Requirement	Requirement summary	CSW capability (IT-side)
Section II	Critical Cyber System identification	Inventory + scope labelling for CSW-managed workloads supporting CCS (§2)
Section II	Cybersecurity Coordinator	Quarterly evidence pack to the Cybersecurity Coordinator
Section III.A	Network segmentation (IT/OT)	Per-site IT-side enclave segmentation (§3); deny-by-default to boundary firewall (§3)
Section III.A	Documented data flows	IT → OT flow inventory + observed-vs-documented delta (§4)
Section III.B	Access control measures	Workload-level least-privilege; quarterly access review (§5)
Section III.C	Continuous monitoring & detection	Behavioural rules + flow telemetry to SIEM (§6)
Section III.D	Patch / unpatched-system risk	Continuous CVE inventory + compensating-control register (§7)
Cybersecurity Incident Response Plan	Detection, response, 24-hour CISA reporting	Six-artefact reconstruction bundle (§8)
Cybersecurity Assessment Plan	Annual implementation assessment	Per-section evidence pack with prior-period delta (§9)
Cybersecurity Architecture Design Review	Periodic architecture review	Workload inventory, dependency graph, flow inventory as inputs (§10)

12. TSA Inspector Response Guide

When TSA inspectors ask for evidence under the SD:

Inspector asks	You provide
"Show your Critical Cyber System inventory and the IT systems that support them"	Per-scope inventory snapshot with <code>ccs_id</code> , <code>ccs_function</code> , <code>ot_facing</code> , <code>site</code> labels (§2)
"Demonstrate your Section III.A IT/OT segmentation"	Workspace export per site + simulation → enforcement log (§3)
"Provide the documented IT-to-OT data flow inventory"	Observed IT → OT flow inventory + reconciliation against architecture (§4)
"Show your Section III.B access control implementation"	Policy export + quarterly review minutes + sunset log (§5)
"Provide Section III.C continuous monitoring evidence"	Detection rule catalogue + SIEM forwarding + MTTD report (§6)
"Show Section III.D patch programme evidence"	CVE inventory dated to period + remediation tracking + compensating-control register (§7)
"Provide the incident reconstruction for event X"	Six-artefact incident dossier (§8)
"Show your CAP for the most recent annual cycle"	Annual CAP evidence binder organised by SD section (§9)

13. Related Frameworks

The TSA Pipeline SD shares substantial control surface with:

- **NIST SP 800-53 Rev 5** — almost all SD substantive requirements have direct 800-53 analogues (AC-3, AC-4, SC-7, SI-4, RA-5, CM-2, CM-3, CM-7, AU-2, IR-4 cover the bulk of Sections III.A–III.D). See [800-53 reference design](#).
- **NIST SP 800-207 (ZTA)** — the deny-by-default, identity-aware, observability-driven posture in §3, §5, and §6 implements the seven ZTA tenets in the pipeline IT context. See [800-207 reference design](#).
- **NIST SP 800-82 Rev 3** (*out of repository scope but worth pairing externally*) — guide to OT security; CSW addresses the IT estate around the OT footprint that 800-82 describes.
- **NERC CIP** — sister sector framework for the bulk electric system; identical IT-side patterns apply. See [NERC CIP reference design](#).
- **ISO/IEC 27001:2022** — entities operating under ISO 27001 can re-use Annex A.8 evidence. See [ISO 27001 reference design](#).
- **CISA Zero Trust Maturity Model** — the SD's substantive requirements align with CISA ZTMM Network and Application/Workload pillars on the IT side. See [CISA ZTMM reference design](#).

14. Disclaimer

This reference design is **draft v1** describing how Cisco Secure Workload capabilities can support a TSA Pipeline Security Directive cybersecurity programme on the **IT side** of the IT/OT boundary. It is **not** legal or regulatory advice and does not constitute a finding of compliance.

Specifically:

- TSA Security Directives are issued and revised by the Transportation Security Administration; the active revision letter (02, 02A, 02B, 02C, 02D, ...) and the substantive requirements evolve. Always validate against the **current effective revision** available from TSA, against your inspector's current guidance, and against any subsequent rulemaking that supersedes the SD.
- Designation as a Critical Pipeline owner/operator is a TSA determination. Nothing in this reference design substitutes for that designation analysis.
- CSW does not enforce policy on RTUs, flow computers, PLCs, IEDs, HMIs, or any other OT device, and is not certified as an OT- protocol deep-packet-inspection or architecture review tool.

This document should receive subject-matter-expert review for both TSA SD regulatory accuracy and current Cisco product capability before being relied upon in a formal compliance engagement.